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Case 10-2023: A 27-Year-Old Man with Convulsions

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PRESENTATION OF CASE

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N Engl J Med 2023;388:1210-8.
DOI: 10.1056/NEJMcpc2211365
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Dr. Nathan C. Praschan (Psychiatry): A 27-year-old man was admitted to this hospital because of convulsions.

The patient had been in his usual state of health until 2 years before this admission, when shaking of the trunk, arms, and legs occurred. The episode lasted approximately 4 to 5 minutes and had been preceded by a sharp pain in the back of the head that radiated to the neck, blurred vision, palpitations, nausea, and difficulty concentrating. The patient did not seek evaluation, and no additional convulsive episodes occurred during the subsequent 20 months.

Four months before this admission, while the patient was drinking alcohol at a party, his body suddenly became stiff, and he fell backward onto a table. Shaking of the trunk, arms, and legs occurred for approximately 5 to 8 minutes. He had urinary incontinence but did not bite his tongue. Five minutes later, similar convulsions occurred. The patient was taken to the emergency department of another hospital for evaluation. The results of computed tomography of the head were normal, as was an electroencephalogram (EEG) that was obtained over a period of 40 minutes. Treatment with levetiracetam was started. The patient was discharged home with a recommendation for follow-up evaluation at the neurology clinic of the other hospital.

Three months before this admission, the patient was evaluated at the neurology clinic of the other hospital. He reported that despite the use of levetiracetam, convulsions occurred approximately once per week. The results of magnetic resonance imaging (MRI) of the head, performed before and after the administration of intravenous contrast material, were reportedly normal. Treatment with lamotrigine was started. There was no change in the quality or frequency of the convulsive episodes.

One month before this admission, while the patient was at work, his colleagues found him on the floor, and they witnessed “shaking and clenching” movements of the trunk, arms, and legs that lasted approximately 2 minutes. Emergency medical services were called, and the patient was taken to the emergency department of a second hospital for evaluation.

At least two additional convulsive episodes occurred at the second hospital. During one episode, the shaking movements became less intense when the physician's hand was placed on the patient's shoulder; the movements resumed at the previous intensity when the hand was removed. In addition, the patient's eyes were closed. When the eyelids were opened manually, no nystagmus or gaze deviation was present. When the eyelids were released, the eyes shut forcefully and Bell's phenomenon (upward deviation of the eyes when closing) occurred. The patient did not speak but was able to follow simple verbal commands, such as an instruction to squeeze the physician's hands. In between the convulsive episodes, there were intermittent rocking movements of the upper body and non-rhythmic tremulous movements of the hands, and the patient did not speak or follow commands. After several hours, the episodes stopped, and the patient was able to speak. Admission to the hospital was recommended; however, the patient declined because of work obligations and anxiety about being in the hospital. He was discharged home.

During the subsequent month, the patient continued to take levetiracetam and lamotrigine. However, the convulsions increased in frequency to one to three times per week.

On the morning before the current admission, while the patient was at work, his colleagues noticed that he abruptly stopped speaking and began to have shaking movements of the arms and legs. Emergency medical services were called. On their arrival, the patient's body was stiff, and there was twitching of the trunk, arms, and legs. No urinary or fecal incontinence or biting of the tongue had occurred. The patient was able to stand up from a chair and walk to the ambulance. He was brought to the emergency department of this hospital.

On examination, the temperature was 36.2°C, the blood pressure 111/56 mm Hg, the pulse 83 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 98% while the patient was breathing ambient air. He was awake and alert but did not speak. His eyes were open, with a primary midline gaze and blinking in response to visual threat. His body appeared stiff, with the arms and legs straightened. There were intermittent nonrhythmic tremulous movements of the trunk, arms, and

legs. The patient was able to follow complex commands and to move the arms and legs symmetrically and against gravity, but there was a delay in motor activation. The deep-tendon reflexes were normal. Extensive scarring from burns was present on the face, trunk, back, arms, and legs.

Approximately 45 minutes after the tremulous movements had stopped, the patient was able to speak but with limited fluency. Additional history was obtained. He had been taking levetiracetam and lamotrigine as directed; however, 3 days before this evaluation, he had stopped taking these medications because he was unsure whether they were effective. That morning, he had felt well until he noticed a sharp pain in the back of his head that radiated to his neck, along with sudden blurred vision, full-body numbness, and difficulty concentrating. A convulsive episode then occurred that lasted up to 20 minutes; during the episode, the patient was aware of events but sometimes had difficulty hearing. He described his overall mood as "pretty good" and did not report decreased energy or chronic fatigue. There was no history of psychotic or manic symptoms, suicidal ideation, or obsessive or compulsive behaviors.

The patient had a history of a febrile seizure in childhood, as well as several concussions that had occurred while he was playing sports in childhood. He reported that during the first 6 years of life, his social supports were limited, and he had experienced maltreatment by a family member; he did not elaborate on the nature of the maltreatment. In late childhood, he was involved in a house fire and hospitalized for a prolonged period because of severe burns on the face, body, arms, and legs. He had asthma, major depressive disorder, chronic post-traumatic stress disorder (PTSD), and panic disorder with agoraphobia. He took escitalopram daily and lorazepam as needed for panic attacks. He was in a long-term romantic relationship. He did not smoke cigarettes or use illicit drugs; he smoked marijuana approximately once per year and drank one alcoholic beverage per week. His father had died of heart disease when the patient was a teenager; his mother had schizoaffective disorder and post-traumatic epilepsy.

Blood levels of glucose and electrolytes were normal, as were the results of liver-function and kidney-function tests. The blood lactate level

was 2.4 mmol per liter (22 mg per deciliter; reference range, 0.5 to 2.2 mmol per liter [5 to 20 mg per deciliter]) and the blood creatine kinase level 489 U per liter (reference range, 60 to 400). Serum and urine toxicologic panels were negative. An electrocardiogram showed normal sinus rhythm. The patient was admitted to this hospital.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. David L. Perez: I participated in the care of this patient, and I am aware of the final diagnosis. In this 27-year-old man, paroxysmal neurologic events with convulsions had occurred since he was 25 years of age. In developing a differential diagnosis, I will consider neurologic, medical, and psychiatric causes. In particular, I will discuss conditions in five broad diagnostic categories: epilepsy, paroxysmal movement disorders, intoxication and withdrawal syndromes, convulsive syncope, and functional neurologic disorder.

EPILEPSY

When the patient was evaluated by clinicians at the first hospital, the leading diagnosis was epilepsy, and treatment with an anticonvulsant drug (levetiracetam) was started. When the patient continued to have convulsions, a second medication (lamotrigine) was added. In considering the diagnosis of epilepsy, some features that are often noted are the duration of the convulsions and whether the patient had urinary incontinence and abnormal laboratory test results.

Most epileptic seizures have a duration of less than 2 minutes, and it would be unusual for epileptic seizures to last 4 to 20 minutes then resolve spontaneously. Status epilepticus — which is characterized by the occurrence of either one epileptic seizure that lasts more than 5 minutes or more than one epileptic seizure within a 5-minute period with incomplete recovery of consciousness in between events — is a condition that reflects a failure of seizure termination.^{1,2} Although an assessment for epilepsy usually involves an inquiry regarding urinary incontinence, this feature cannot be used to reliably differentiate epileptic seizures from syncope or functional seizures.³

After convulsive epileptic seizures, the peak creatine kinase level is typically 8 to 19 times as high as the upper limit of the normal range^{4,5}; in this patient, there was only a modest increase in

the creatine kinase level. Similarly, a postictal lactate level of 2.9 mmol per liter (26 mg per deciliter) or higher has been reported to have 82% specificity in differentiating epileptic seizures from functional seizures⁶; in this patient, the lactate level was below that threshold.

A routine EEG that was obtained while the patient was asymptomatic was normal; however, acquisition of a single EEG has only modest sensitivity (20 to 50%) for the detection of interictal epileptiform discharges.⁷ As such, the available EEG data in this patient cannot rule out the possibility of epilepsy. Although I am not yet ready to rule out epileptic seizures, I have a low level of suspicion for this diagnosis, given the long duration of the events and the subthreshold creatine kinase and lactate levels, as well as the normal MRI results and the semiologic features of the events.

PAROXYSMAL MOVEMENT DISORDERS

This patient had convulsive events with an element of apparent preserved wakefulness that could suggest a paroxysmal movement disorder.^{8,9} Paroxysmal movement disorders are rare and include paroxysmal kinesigenic dyskinesia, paroxysmal nonkinesigenic dyskinesia, paroxysmal exercise-induced dyskinesia, episodic ataxias, and other conditions. In patients with paroxysmal kinesigenic dyskinesia, events typically begin to occur before 20 years of age, are precipitated by a movement-based trigger, have a duration of less than 1 minute, and are associated with stereotyped movements.⁸ In patients with paroxysmal nonkinesigenic dyskinesia, events usually begin to occur in infancy or early childhood, can be precipitated by a trigger that is not movement-based (e.g., caffeine or alcohol use), have a duration of minutes to hours, and are associated with stereotyped movements.⁸ The spectrum of paroxysmal dyskinesia extends beyond these classic descriptions, but most conditions in this category are inherited.⁹ In this patient, the absence of a family history of similar episodes (his mother's epilepsy was post-traumatic), the semiologic variability, and the onset of symptoms after 20 years of age make these disorders unlikely.

INTOXICATION AND WITHDRAWAL SYNDROMES

The possibility that the patient's symptoms were related to substance use is worth considering. At least one event occurred while he was drinking alcohol. Furthermore, events had been preceded

by headache, palpitations, and nausea. The patient had received a prescription for lorazepam to be taken as needed, and he had left another emergency department early because of anxiety; these aspects of his history suggest the possibility of benzodiazepine misuse. Intoxication with certain drugs of abuse (e.g., cocaine) and withdrawal from others (e.g., alcohol or benzodiazepines) have been associated with epileptic seizures.¹⁰ However, this patient did not have ongoing abnormalities — such as tachycardia, diaphoresis, or pupillary findings — that would support the presence of an acute intoxication or withdrawal syndrome. The negative serum and urine toxicologic panels are also not consistent with substance use.

CONVULSIVE SYNCOPE

Because convulsions in this patient had been preceded by palpitations, cardiac-related convulsive syncope is also worth considering. However, the motor symptoms in this patient lasted several minutes, whereas the motor phenomena in patients with convulsive syncope usually last less than a minute.¹¹ The normal electrocardiogram further argues against the diagnosis of convulsive syncope.

FUNCTIONAL NEUROLOGIC DISORDER

I will now consider whether this patient's symptoms are consistent with functional seizures, a common subtype of functional neurologic disorder.¹²⁻¹⁴ Functional seizures are also known as psychogenic nonepileptic seizures or dissociative seizures. The following semiologic features that were described by emergency department physicians and consulting neurologists in this case are particularly notable: the closed eyes and forceful shutting of the eyelids that occurred during a convulsive event, the pattern of non-rhythmic tremulous movements, and the decrease in the intensity of shaking movements when the physician placed a hand on the patient's shoulder, which indicated a modification of the convulsive event by interaction with another person.¹³

In a meta-analysis of 14 studies that involved 800 patients with epileptic seizures and 452 patients with functional seizures, the presence of ictal eye closure and asynchronous limb movements had a moderate-to-large positive effect on the probability of functional seizures.¹⁵ In a prospective study, the ability of bystanders to

alleviate or intensify seizures had a high specificity (>80%) for functional seizures, as compared with epileptic seizures.¹⁶ In addition, when the ictal duration is 5 minutes or longer, the odds of functional seizures are approximately 24 times as high as the odds of epileptic seizures.¹

Another notable feature in this case is that events were preceded by additional physical symptoms, including headache, palpitations, nausea, and difficulty concentrating. In patients with functional seizures, a phenomenon characterized as “panic attack without panic” has been described; functional seizures can be preceded by or occur with autonomic and arousal symptoms that are associated with panic attacks, sometimes without the emotional experience of panic.¹⁷⁻¹⁹

The patient's history of psychiatric conditions and childhood maltreatment does not feature prominently in my consideration of the differential diagnosis. Mental health conditions are common among patients with functional seizures and among patients with epileptic seizures.^{20,21} A history of childhood maltreatment is common among patients with functional neurologic disorder but is not always present.²² Furthermore, approximately one in five patients with functional seizures also has epileptic seizures, so careful consideration of the semiologic features in each patient is warranted.²³

Taken together, this patient's clinical history, semiologic features, and available EEG data meet the criteria for a probable diagnosis of functional seizures, as defined in a consensus statement by the International League Against Epilepsy (Table 1).²⁴ Given the high event frequency, I would recommend inpatient admission for continuous video EEG monitoring to capture clinical events in order to provide additional diagnostic certainty.

DR. DAVID L. PEREZ'S DIAGNOSIS

Functional seizures.

DIAGNOSTIC TESTING

Dr. Daniel B. Hoch: The patient was admitted for continuous video EEG monitoring to be performed over a period of 3 days. For patients who are already taking anticonvulsant drugs, inpatient admission is preferred to allow for safe tapering of the drugs and immediate availability

Table 1. Diagnostic Criteria for Functional Seizures According to Level of Diagnostic Certainty.*

Level of Diagnostic Certainty	Clinical History†	Description of Seizure	Available EEG Data	Present in This Patient‡
Possible	+	A nonclinical witness or the patient describes possible semiologic features typical of functional seizures.	Routine or sleep-deprived interictal EEG shows no epileptiform activity.	—
Probable	+	A clinician (who reviewed a video recording or observed the event in person) describes semiologic features typical of functional seizures.	Routine or sleep-deprived interictal EEG shows no epileptiform activity.	Before admission
Clinically established	+	A clinician with experience in the diagnosis of seizure disorders (who reviewed a video recording or observed the event in person) describes semiologic features typical of functional seizures.	Routine or ambulatory ictal EEG that captured a typical seizure event shows no epileptiform activity.§	—
Documented	+	A clinician with experience in the diagnosis of seizure disorders (who reviewed a video recording or observed the event in person) describes semiologic features typical of functional seizures.	Ictal video EEG shows no epileptiform activity immediately before, during, or after the seizure event but shows semiologic features typical of functional seizures.	After video EEG

* Data are from LaFrance et al.²⁴ EEG denotes electroencephalogram.

† The plus signs indicate a clinical history consistent with functional seizures.

‡ Semiologic features typical of functional seizures that were present in this patient were ictal eye closure, asynchronous limb movements, a long duration of events, and alleviation of the intensity of the seizure by a bystander. Urinary incontinence and tongue biting cannot be used to reliably differentiate epileptic seizures from functional seizures.

§ The captured event cannot resemble types of epileptic seizures that may not be correlated with epileptiform activity on surface EEG.

of staff during clinical events. This patient reported that he had stopped taking anticonvulsant drugs 3 days before the admission.

During the admission, the patient's typical symptoms occurred three times. During the first episode, the patient noticed being awakened by a noise and looking at the cardiac monitor. The nurse who entered the room noticed that the patient appeared confused but was able to follow a verbal command to point to the window; when he was asked to point to the door, he did not respond. The second episode occurred with photic stimulation. When the patient was exposed to a flickering light, he closed his eyes, moved his head from left to right, and became unresponsive. After 3 minutes, he reported that he was "coming out of a seizure." The third episode occurred on the final day of monitoring. The patient pushed the alarm button and reported that he was feeling shaky. When the nurse asked if he was having his typical convulsive symptoms, he said "yes." The patient then stopped

interacting with the nurse and the physician who had entered the room. He closed his eyes, raised his right hand to touch his nose, and kept his hand in place. When the physician tried to gently raise the patient's eyelids, Bell's phenomenon occurred. The continuous video EEG recording remained normal during the three episodes of symptoms (Fig. 1), a finding that confirmed the diagnosis of functional seizures.

CLINICAL DIAGNOSIS

Functional seizures.

DISCUSSION OF MANAGEMENT

Dr. Perez: Treatment with anticonvulsant drugs was discontinued, education on the diagnosis of functional seizures was provided, and the patient was discharged home with a plan for follow-up visits in the functional neurologic disorder clinic of this hospital. Outpatient management focused

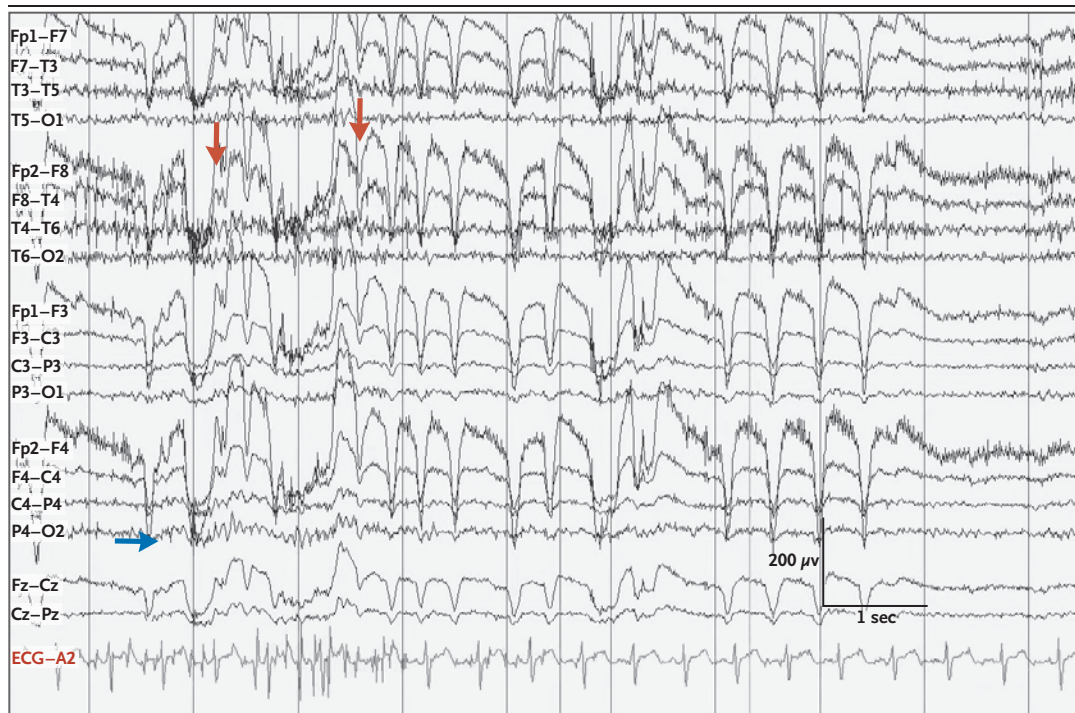


Figure 1. EEG Obtained During a Convulsive Episode.

This electroencephalogram (EEG) was obtained over a period of 10 seconds during a convulsive episode. During the episode, multiple eye movements occurred. There was a brief period of eye closure (first red arrow), which correlates with a normal, 11-Hz posterior-dominant rhythm (blue arrow). The period of eye closure was followed by multiple eye blinks (second red arrow), which correlate with attenuation of the posterior-dominant rhythm. No epileptiform features were noted at any time during the episode.

on education, engagement, continued assessment for psychiatric and psychosocial factors, the administration of medications for coexisting conditions, and a referral for psychotherapy.^{25,26}

During the first follow-up visit, the patient was guarded, with minimal spontaneous endorsement of symptoms or concerns. On direct questioning, he reported three seizures that had occurred in the previous month, along with depression, agoraphobia, and panic attacks that had preceded the seizures. He reported no PTSD symptoms. An attempted transition from escitalopram to sertraline, which had been recommended by his primary care physician, had led to adverse effects, so he continued to take escitalopram.

The patient and I discussed that functional seizures are at the intersection of neurology and psychiatry. They are real and common, and treatments are available, although this condition is not treated with anticonvulsant drugs.^{13,27,28} We discussed some mechanistic analogies for

functional seizures; they have been described as a software rather than a hardware problem and as a form of brain–mind–body overload.^{29,30}

Bupropion therapy was started as additional treatment for major depression. Psychotherapy was suggested so that we could learn more about and address the patient’s “personal equation” for the development and maintenance of functional seizures.³¹ However, he expressed ambivalence about this referral, citing financial concerns. Although the patient was initially guarded during the visits, he called the clinic between visits on several occasions to report episodes of functional seizures and worsening depression without suicidality.

Eleven months after discharge, the patient readily endorsed PTSD symptoms, including nightmares, intrusive daytime recollections, avoidance, and prominent dissociation.^{32,33} Prazosin therapy was started for the treatment of PTSD-related nightmares. At this time, the patient agreed to pursue psychotherapy at the

Table 2. Management Strategies for Functional Seizures.

Strategies	Considerations	Resources
Education	Communicate the diagnosis in a transparent and empathetic manner with ample time for discussion. Provide resources.	Neurosymptoms.org Carson et al. ²⁷ Hall-Patch et al. ³⁰
Engagement	Anticipate potential ambivalence about treatment. Use validation, empathetic listening, and motivational interviewing strategies.	Tolchin et al. ²⁶
Psychotherapy	Consider cognitive behavioral therapy, prolonged exposure therapy, mindfulness-based therapy, dialectical behavioral therapy, acceptance and commitment therapy, short-term dynamic psychotherapy, or a combination thereof.	Gutkin et al. ³¹
Treatment of coexisting conditions	Do not use anticonvulsant drugs unless there is a concern for coexisting epileptic seizures or another indication is present. Consider pharmacologic management of decompensated mental health conditions (e.g., post-traumatic stress disorder) and medical or neurologic conditions (e.g., migraines).	Perez and LaFrance ¹²
Longitudinal neurologic and mental health follow-up	Prevent dualistic impressions of the diagnosis that may occur with poorly coordinated neurologic and mental health follow-up. Encourage multidisciplinary integrated clinical neuroscience approaches, in which neurologists or neuropsychiatrists work in collaboration with psychotherapists longitudinally.	Lopez and LaFrance ³³
Other biopsychosocial considerations	Evaluate for predisposing vulnerabilities, acute precipitants, and perpetuating factors. Identify case-specific factors that require special attention in the treatment plan, such as active symptoms of post-traumatic stress disorder.	Finkelstein et al. ²⁵

PTSD clinic of this hospital, recognizing that PTSD was an identifiable perpetuating factor for his functional seizures. Specific forms of psychotherapy that are often used for the treatment of functional seizures, such as cognitive behavioral therapy,^{34,35} were considered. However, this patient's biopsychosocial clinical formulation suggested that a more personalized approach was most likely to have a positive outcome at this early stage in his treatment (Table 2).³⁶

PSYCHOTHERAPY

Dr. Jonah N. Cohen: Psychotherapeutic theories and treatments for functional neurologic disorder have evolved from the late 19th and early 20th century writings of Freud and Breuer.³⁷ In addition to the theory that repression is a potential underlying mechanism, there are more recent theories, including disturbances related to emotional regulation, dissociation, and attachment, as well as underlying trauma and cultural considerations. Cognitive and dialectical behavioral therapies can be helpful; however, there is no single or algorithmic approach to the treatment of functional seizures. Because of the complexity and heterogeneity of functional seizures, an

integrative and developmentally informed approach to treatment is most effective.³⁸

COMPLEX TRAUMA

This patient's functional seizures (a condition that can go by several different names³⁹) could be understood as a symptom of complex trauma. Complex trauma refers to trauma that is often repeated and is associated with emotional regulation difficulties, interpersonal challenges, and a limited capacity to cope with life stressors.⁴⁰ Persons with complex trauma tend to have a fragmented sense of self ("me-ness"). Different parts of themselves feel disconnected. Such disconnection develops when certain experiences, such as trauma reminders, are dissociated, which means that they are pushed out of awareness because they are too threatening. By contrast, persons without complex trauma tend to have a more intact sense of self and are less likely to use dissociation. Different parts of themselves, even parts that are upsetting, feel more fundamentally linked.

Complex trauma can be treated with a phase-based approach that focuses on establishing a sense of safety and stabilization before moving

on to directly address traumatic memories.^{41,42} In addition to pharmacotherapy, the psychotherapeutic relationship is foundational to successful treatment.⁴³ The long-term care of a patient with complex trauma can be challenging, and a strong psychotherapeutic relationship can teach patients what a trusting relationship — one that has the capacity to withstand mistrust and anger — feels like.

The first phase of this patient's treatment focused on learning about the nature of his PTSD, dissociation, and functional seizures. We discussed that his experiences are real and not "all in his head." Psychoeducation is crucial when working with patients who have received a diagnosis of functional seizures, because the seizures are often painful, confusing, and socially stigmatic.

Next, we worked on developing a language to map his emotions, including sadness, fear, and anger. The patient was not aware of what it felt like to dissociate, nor was he aware of any affective precursors to his dissociative disturbances, so we analyzed his experiences of dissociation and their antecedents.

Once the patient had the tools to identify elements of his internal experiences, he began to learn tools to regulate those experiences. This phase of his treatment was informed by dialectical behavioral therapy. He learned to recognize and then leave a tense situation and to use distraction and diaphragmatic breathing. We practiced these skills repeatedly when he was in various emotional or arousal states, which ranged from being calm to being on the brink of dissociation.

Next, the patient learned cognitive strategies to help regulate his emotions in more adaptive ways. Learning about cognitive distortions such as catastrophizing (predicting that the worst will happen) and mind reading (thinking that you know what is going on in someone else's mind) taught him that thoughts are not always true. We also used exposures to address triggers to his anxiety.

Gradually, the patient had an increased ability

to be aware of and then regulate his experiences. Our psychotherapeutic relationship strengthened, and he requested a referral to the burn clinic at this hospital; he had been avoiding an evaluation to further treat the injuries that he had sustained in the house fire.

In summary, treatment of this patient required an empirical approach, rather than a more manualized approach. The patient had experienced isolation and abuse early in his life and had not learned how to symbolize or regulate certain experiences; instead, upsetting experiences were dissociated, potentially because the emergence of traumatic memories into consciousness threatened his sense of being. The most distressing trauma reminders, and high arousal states more broadly, triggered seizures.

The patient learned strategies to identify and regulate challenging emotional experiences — to formulate the unformulated⁴⁴ — and our psychotherapeutic relationship was critical to this process. During our sessions, I witnessed the patient's experiences and reflected them back to him; he came to know more about himself through the process of my coming to know him.⁴⁵ The patient is learning to confront an internal and external world that is largely unfathomable to me, yet perhaps gradually increasingly fathomable to him.

FOLLOW-UP

Dr. Cohen: The patient has continued his neuropsychiatric care and psychotherapy. He remains in a committed relationship and has started a full-time job. Nevertheless, he continues to experience dissociation and isolation, and he still has some, albeit fewer, functional seizures.

FINAL DIAGNOSIS

Functional seizures.

This case was presented at Psychiatry Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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